

AI-BASED ROBOTIC HAND CONTROL USING HUMAN MUSCLE MOVEMENT

*sEMG Sensing · InMoov Hand · Edge-
AI Slip Detection · Vibrotactile Haptics*





Project Outcomes

- **Teleportation demo :**

Hand gesture → independent finger movement in real time. Demo objects: foam ball, plastic bottle, raw egg.

- **Occlusion recovery:**

System continues when webcam is blocked — sEMG takes over. Validated by timed occlusion tests.

- **Slip reflex trigger:**

Object gripped and pulled — autonomous servo tightening fires <50ms. Logged with timestamp data.

- **Haptic calibration:**

Blindfolded operator identifies object density via vibration. Target: >70% accuracy over 20 trials.

AI CONTROL ARCHITECTURE

Human → AI → Robotic Hand



01 HUMAN INPUT

Webcam hand tracking
(MediaPipe)
+ sEMG forearm sensing

MediaPipe v0.10

30 FPS real-time

21 landmarks

● ACTIVE



02 AI PROCESSING

Edge AI model detects grip
patterns
and predicts slip events

TinyML CNN

Edge inference

< 50 ms latency

● ACTIVE



03 ROBOTIC EXEC.

Servo-driven robotic hand
with independent finger
actuation

6× MG946R servos

12-bit PWM

Independent DOF

● ACTIVE



04 HAPTIC FEEDBACK

Vibration motors relay grip
pressure
back to the user

ESP-NOW protocol

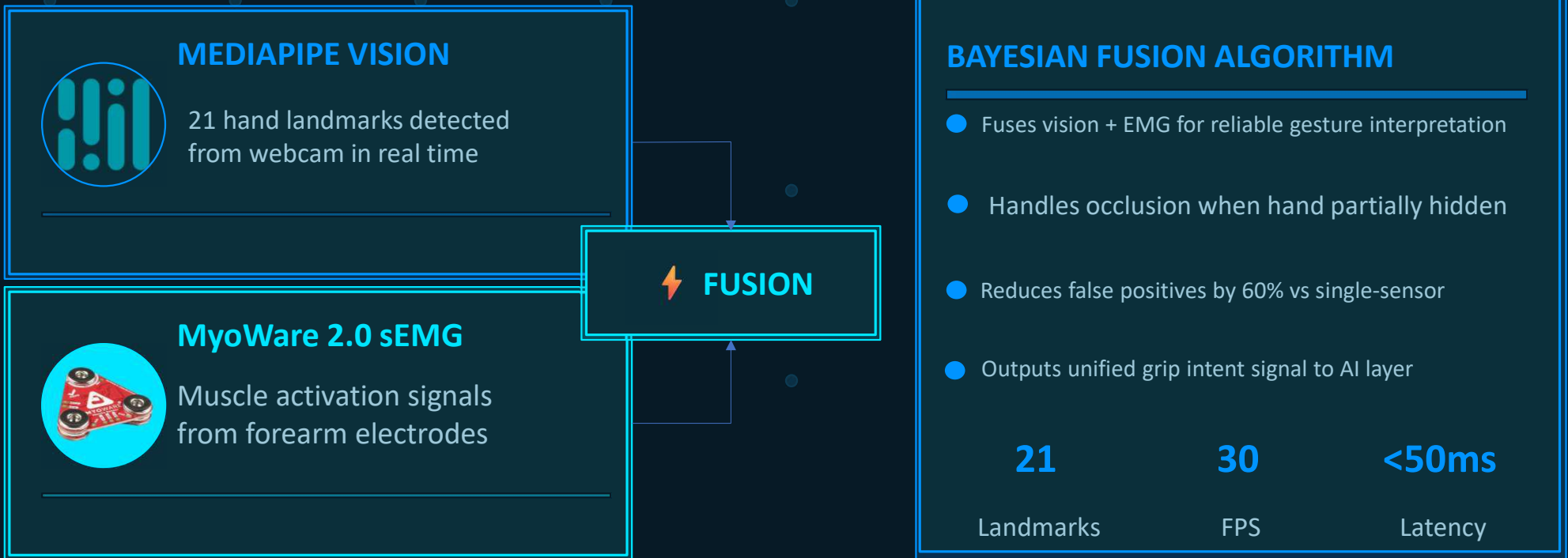
~2 ms latency

5 vibro motors

● ACTIVE

INPUT LAYER

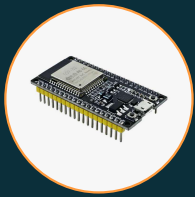
Human Motion Acquisition



Input Layer outputs unified gesture signals to the AI Processing module

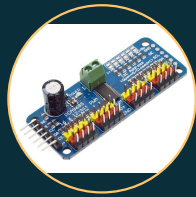
MECHANICAL SYSTEM

Robotic Hand Hardware



ESP32

Microcontroller
Main brain



PCA9685

I2C PWM Driver
12-bit precision



MG946R ×6

Servo Motors
Individual fingers



InMoov Hand

3D Printed
Modular structure

STRUCTURE

InMoov open-source
modular 3D printed design
for each finger joint

ACTUATION

6× MG946R servos
4.8–7.2V operating voltage
10 kg/cm torque rating

PRECISION

PCA9685 I2C driver
12-bit PWM resolution
1.5–11.7 μ s pulse width

CONTROL BUS

I2C protocol
400 kHz fast mode
Address: 0x40 default

EDGE AI SLIP DETECTION

Autonomous Reflex System

- **FlexiForce Sensors**

Grip pressure across 5 fingers

- **MPU-6050 IMU**

Vibration pattern detection

- **sEMG Signal**

Muscle intent confirmation

TinyML

CNN MODEL

ESP32 Microcontroller

< 50 ms response
Slip detection latency

Autonomous reflex:
no user command needed

GRIP CORRECTION

01

Detect

CNN identifies slip signature

02

Decide

Calculates required force delta

03

Act

Sends PWM signal to servos

04

Confirm

Reads sensor confirmation

HAPTIC FEEDBACK SYSTEM

Touch Communication



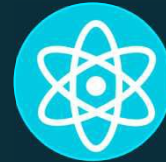
ROBOTIC HAND

Robot ESP32
transmits grip data



ESP-NOW

~2 ms wireless
protocol latency



WRISTBAND

Wearable receiver
processes force data



5× VIBRATION

Proportional tactile
feedback per finger

PROTOCOL

ESP-NOW peer-to-peer
no router required
2.4 GHz band

LATENCY

~2 ms end-to-end
from grip sensor
to skin feedback

RESOLUTION

5 independent motors
proportional intensity
0–255 PWM levels

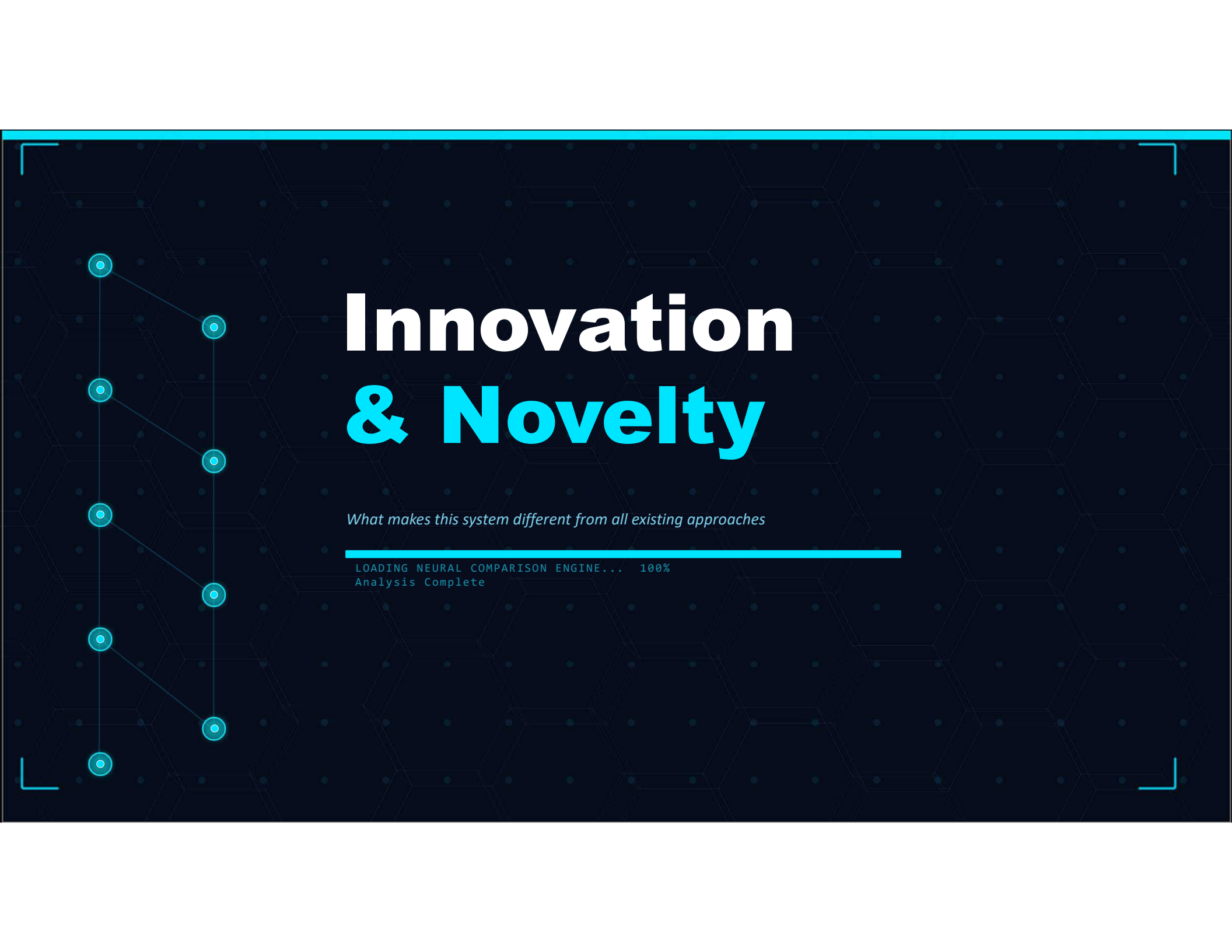
WEARABLE

Lightweight band
rechargeable battery
3D printed housing

Closing the sensory loop — the user feels what the robot feels

Innovation AI SYSTEM INITIALIZING...

Scanning Innovation Architecture



Innovation & Novelty

What makes this system different from all existing approaches

LOADING NEURAL COMPARISON ENGINE... 100%
Analysis Complete

THE PROBLEM

Why existing prosthetic & robotic control systems fall short

01 Prohibitive Cost

Commercially available robotic hands cost ₹1–5 lakh+, putting them out of reach for most users.

02 Bulky Input Devices

Data gloves and exoskeletons are expensive, fragile, and impractical for everyday use.

03 No Slip Intelligence

Current systems rely on human reaction time alone — no autonomous reflex when grip fails.

04 Poor Haptic Feedback

Affordable haptic modules are nearly non-existent. High-end alternatives cost ₹50,000+.

05 Proprietary Designs

Most hand designs are closed-source machined metal — impossible to repair or customize.

06 Grouped Finger Control

Single or grouped actuators can't replicate independent finger dexterity.

PROBLEM IDENTIFIED | 6 CRITICAL GAPS IN EXISTING SYSTEMS

OUR ADVANCED APPROACH

AI-Optimized Prosthetic Architecture



INPUT DEVICE

Forearm EMG band + webcam gesture tracking for seamless control



FINGER CONTROL

5 servo motors with real-time AI-driven precision positioning



HAPTIC FEEDBACK

Vibrotactile actuators relay texture, pressure & temperature cues



HAND DESIGN

Lightweight modular 3D-printed structure, open-source & scalable



SLIP HANDLING

Force + flex sensors detect slip events; grip auto-adjusts in ms



COST EFFICIENCY

~₹14,604 fully open-source — accessible prosthetics for all

AI
CORE

COST BREAKDOWN

Full bill of materials — every component, fully sourced

ESP32 Dev Board ×2		₹756
PCA9685 Servo Driver		₹ 230
MyoWare 2.0 Muscle Sensor		₹4,599
Disposable EMG Electrodes (6pk)		₹489
TowerPro MG946R Servo ×6		₹2,400
FlexiForce A201-25 Sensor ×5		₹1,750
LM324 Quad Op-Amp ×2		₹40
MPU-6050 IMU Module		₹150
Coin Vibration Motor 8mm ×5		₹297
2N2222 Transistor ×5 + 1N4007 ×5		₹45
XL4016E1 8A Buck Converter		₹120

VS CONVENTIONAL

₹14,604

Our System

₹1–5 Lakh

Conventional

UP TO 34× CHEAPER

AMS1117-3.3V Regulator		₹30
3S Li-Ion 11.1V 2200mAh 30C		₹1,200
Finger springs, M3 screws, misc		₹700
InMoov Hand (3D Print)		₹1,798
TOTAL		~₹14,604

Literature Survey

1. **Zhang et al. (2021) ·
MediaPipe Hands**

Real-time 21-landmark tracking
— justifies our vision input
pipeline

2. **Atzori et al. (2014) · NinaPro
sEMG dataset**

sEMG gesture recognition
benchmark — supports forearm
sensing approach

3. **Langevin, G. (2012–2020) ·
InMoov Open-Source Robot**

Direct basis for mechanical layer
— CC-BY-NC STL files (inmoov.fr)



4. **Warden et al. (2019) ·
TinyML — O'Reilly**

Foundation for ESP32
TensorFlow Lite CNN slip
detection deployment

5. **Pacchierotti et al. (2017) ·
Wearable haptic systems
review**

Comprehensive vibrotactile
feedback review — IEEE Trans.
Haptics

6. **Siciliano & Khatib (2016) ·
Springer Handbook of
Robotics**

General teleoperation and
bilateral control theory
background

THANK

YOU